1. Write a query to get unique department ID from employee table.
Ans :

```
D3_96509_Aditya> SELECT DISTINCT  DEPARTMENT_ID FROM employees;
+---------------+
| DEPARTMENT_ID |
+---------------+
|          NULL |
|            10 |
|            20 |
|            30 |
|            40 |
|            50 |
|            60 |
|            70 |
|            80 |
|            90 |
|           100 |
|           110 |
+---------------+
12 rows in set (0.00 sec)
```

2. Write a query to get all the employee’ s details from the employees table displaying the first names in descending order.

```
D3_96509_Aditya> SELECT FIRST_NAME FROM employees ORDER BY FIRST_NAME DESC;
+-------------+
| FIRST_NAME  |
+-------------+
| Winston     |
| William     |
| William     |
| Vance       |
| Valli       |
| Trenna      |
| TJ          |
| Timothy     |
| Tayler      |
| Susan       |
| Sundita     |
| Sundar      |
| Steven      |
| Steven      |
| Stephen     |
| Sigal       |
| Shelli      |
| Shelley     |
| Shanta      |
| Sean        |
| Sarath      |
| Sarah       |
| Samuel      |
| Renske      |
| Randall     |
| Randall     |
| Peter       |
| Peter       |
| Payam       |
| Patrick     |
| Pat         |
| Oliver      |
```

3. Write a query to get the employee ID, names (first_name and last_name), salary in ascending order of salary.

```
D3_96509_Aditya> SELECT  EMPLOYEE_ID, FIRST_NAME, LAST_NAME,SALARY FROM employees ORDER BY SALARY ASC;
+-------------+-------------+-------------+----------+
| EMPLOYEE_ID | FIRST_NAME  | LAST_NAME   | SALARY   |
+-------------+-------------+-------------+----------+
|         132 | TJ          | Olson       |  2100.00 |
|         128 | Steven      | Markle      |  2200.00 |
|         136 | Hazel       | Philtanker  |  2200.00 |
|         127 | James       | Landry      |  2400.00 |
|         135 | Ki          | Gee         |  2400.00 |
|         119 | Karen       | Colmenares  |  2500.00 |
|         131 | James       | Marlow      |  2500.00 |
|         140 | Joshua      | Patel       |  2500.00 |
|         144 | Peter       | Vargas      |  2500.00 |
|         182 | Martha      | Sullivan    |  2500.00 |
|         191 | Randall     | Perkins     |  2500.00 |
|         118 | Guy         | Himuro      |  2600.00 |
|         143 | Randall     | Matos       |  2600.00 |
|         198 | Donald      | OConnell    |  2600.00 |
|         199 | Douglas     | Grant       |  2600.00 |
|         126 | Irene       | Mikkilineni |  2700.00 |
|         139 | John        | Seo         |  2700.00 |
|         117 | Sigal       | Tobias      |  2800.00 |
|         130 | Mozhe       | Atkinson    |  2800.00 |
|         183 | Girard      | Geoni       |  2800.00 |
|         195 | Vance       | Jones       |  2800.00 |
|         116 | Shelli      | Baida       |  2900.00 |
|         134 | Michael     | Rogers      |  2900.00 |
|         190 | Timothy     | Venzl       |  2900.00 |
|         187 | Anthony     | Cabrio      |  3000.00 |
|         197 | Kevin       | Feeney      |  3000.00 |
|         115 | Alexander   | Khoo        |  3100.00 |
|         142 | Curtis      | Davies      |  3100.00 |
|         181 | Jean        | Fleaur      |  3100.00 |
|         196 | Alana       | Walsh       |  3100.00 |
|         125 | Julia       | Nayer       |  3200.00 |
|         138 | Stephen     | Stiles      |  3200.00 |
|         153 | Christopher | Olsen       |  8000.00 |
|         159 | Lindsey     | Smith       |  8000.00 |
|         110 | John        | Chen        |  8200.00 |
|         121 | Adam        | Fripp       |  8200.00 |
|         206 | William     | Gietz       |  8300.00 |
|         177 | Jack        | Livingston  |  8400.00 |
|         176 | Jonathon    | Taylor      |  8600.00 |
|         175 | Alyssa      | Hutton      |  8800.00 |
|         103 | Alexander   | James       |  9000.00 |
|         109 | Daniel      | Faviet      |  9000.00 |
|         152 | Peter       | Hall        |  9000.00 |
|         158 | Allan       | McEwen      |  9000.00 |
|         151 | David       | Bernstein   |  9500.00 |
|         157 | Patrick     | Sully       |  9500.00 |
|         163 | Danielle    | Greene      |  9500.00 |
|         170 | Tayler      | Fox         |  9600.00 |
|         150 | Sean        | Tucker      | 10000.00 |
|         156 | Janette     | King        | 10000.00 |
|         169 | Harrison    | Bloom       | 10000.00 |
|         204 | Hermann     | Brown       | 10000.00 |
|         149 | Eleni       | Zlotkey     | 10500.00 |
|         162 | Clara       | Vishney     | 10500.00 |
|         114 | Den         | Li          | 11000.00 |
|         148 | Gerald      | Cambrault   | 11000.00 |
|         174 | Ellen       | Abel        | 11000.00 |
|         168 | Lisa        | Ozer        | 11500.00 |
|         147 | Alberto     | Errazuriz   | 12000.00 |
|         108 | Nancy       | Gruenberg   | 12008.00 |
|         205 | Shelley     | Higgins     | 12008.00 |
|         201 | Michael     | Martinez    | 13000.00 |
|         146 | Karen       | Partners    | 13500.00 |
|         145 | John        | Singh       | 14000.00 |
|         101 | Neena       | Yang        | 17000.00 |
|         102 | Lex         | Garcia      | 17000.00 |
|         100 | Steven      | King        | 24000.00 |
+-------------+-------------+-------------+----------+
107 rows in set (0.00 sec)
```

4. Display first name and joining date of the employees who are either IT Programmer or Sales Man.

```
D3_96509_Aditya> SELECT FIRST_NAME, JOB_ID
    -> FROM employees
    -> WHERE (JOB_ID = 'IT_PROG' OR JOB_ID = 'SA_MAN');
+-------------+---------+
| FIRST_NAME | JOB_ID  |
+-------------+---------+
| Alexander  | IT_PROG |
| Bruce      | IT_PROG |
| David      | IT_PROG |
| Valli      | IT_PROG |
| Diana      | IT_PROG |
| John       | SA_MAN  |
| Karen      | SA_MAN  |
| Alberto    | SA_MAN  |
| Gerald     | SA_MAN  |
| Eleni      | SA_MAN  |
+-------------+---------+
10 rows in set (0.00 sec)

D3_96509_Aditya>
```

5. Display details of employees with employee ID 150 or 160.

```
D3_96509_Aditya> SELECT * FROM employees WHERE (EMPLOYEE_ID = 150 OR EMPLOYEE_ID= 160);
+-------------+------------+-----------+---------+----------------+------------+--------+----------+----------------+------------+---------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL   | PHONE_NUMBER   | HIRE_DATE  | JOB_ID | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+---------+----------------+------------+--------+----------+----------------+------------+---------------+
|         150 | Sean       | Tucker    | STUCKER | 44.1632.960005 | 2015-01-30 | SA_REP | 10000.00 |           0.30 |        145 |            80 |
|         160 | Louise     | Doran     | LDORAN  | 44.1632.960015 | 2015-12-15 | SA_REP |  7500.00 |           0.30 |        146 |            80 |
+-------------+------------+-----------+---------+----------------+------------+--------+----------+----------------+------------+---------------+
2 rows in set (0.00 sec)

D3_96509_Aditya>
```

6. Display first name, salary, commission pct, and hire date for employees with salary less than 10000.

```
D3_96509_Aditya> SELECT FIRST_NAME, SALARY,COMMISSION_PCT,HIRE_DATE FROM emp
loyees WHERE SALARY < 10000;
+-------------+---------+----------------+------------+
| FIRST_NAME  | SALARY  | COMMISSION_PCT | HIRE_DATE  |
+-------------+---------+----------------+------------+
| Alexander   | 9000.00 |           NULL | 2016-01-03 |
| Bruce       | 6000.00 |           NULL | 2017-05-21 |
| David       | 4800.00 |           NULL | 2015-06-25 |
| Valli       | 4800.00 |           NULL | 2016-02-05 |
| Diana       | 4200.00 |           NULL | 2017-02-07 |
| Daniel      | 9000.00 |           NULL | 2012-08-16 |
| John        | 8200.00 |           NULL | 2015-09-28 |
| Ismael      | 7700.00 |           NULL | 2015-09-30 |
| Jose Manuel | 7800.00 |           NULL | 2016-03-07 |
| Luis        | 6900.00 |           NULL | 2017-12-07 |
| Alexander   | 3100.00 |           NULL | 2013-05-18 |
| Shelli      | 2900.00 |           NULL | 2015-12-24 |
| Sigal       | 2800.00 |           NULL | 2015-07-24 |
| Guy         | 2600.00 |           NULL | 2016-11-15 |
| Karen       | 2500.00 |           NULL | 2017-08-10 |
| Matthew     | 8000.00 |           NULL | 2014-07-18 |
| Adam        | 8200.00 |           NULL | 2015-04-10 |
| Payam       | 7900.00 |           NULL | 2013-05-01 |
| Shanta      | 6500.00 |           NULL | 2015-10-10 |
| Kevin       | 5800.00 |           NULL | 2017-11-16 |
| Julia       | 3200.00 |           NULL | 2015-07-16 |
| Irene       | 2700.00 |           NULL | 2016-09-28 |
| James       | 2400.00 |           NULL | 2017-01-14 |
| Steven      | 2200.00 |           NULL | 2018-03-08 |
| Laura       | 3300.00 |           NULL | 2015-08-20 |
| Mozhe       | 2800.00 |           NULL | 2015-10-30 |
| James       | 2500.00 |           NULL | 2015-02-16 |
| TJ          | 2100.00 |           NULL | 2017-04-10 |
| Jason       | 3300.00 |           NULL | 2014-06-14 |
| Michael     | 2900.00 |           NULL | 2016-08-26 |
| Ki          | 2400.00 |           NULL | 2017-12-12 |
| Hazel       | 2200.00 |           NULL | 2018-02-06 |
| Renske      | 3600.00 |           NULL | 2013-07-14 |
| Stephen     | 3200.00 |           NULL | 2015-10-26 |
```

```
| Mattea       | 7200.00 |           0.10 | 2018-01-24 |
| David        | 6800.00 |           0.10 | 2018-02-23 |
| Sundar       | 6400.00 |           0.10 | 2018-03-24 |
| Amit         | 6200.00 |           0.10 | 2018-04-21 |
| Tayler       | 9600.00 |           0.20 | 2016-01-24 |
| William      | 7400.00 |           0.15 | 2017-02-23 |
| Elizabeth    | 7300.00 |           0.15 | 2017-03-24 |
| Sundita      | 6100.00 |           0.10 | 2018-04-21 |
| Alyssa       | 8800.00 |           0.25 | 2015-03-19 |
| Jonathon     | 8600.00 |           0.20 | 2016-03-24 |
| Jack         | 8400.00 |           0.20 | 2016-04-23 |
| Kimberely    | 7000.00 |           0.15 | 2017-05-24 |
| Charles      | 6200.00 |           0.10 | 2018-01-04 |
| Winston      | 3200.00 |           NULL | 2016-01-24 |
| Jean         | 3100.00 |           NULL | 2016-02-23 |
| Martha       | 2500.00 |           NULL | 2017-06-21 |
| Girard       | 2800.00 |           NULL | 2018-02-03 |
| Nandita      | 4200.00 |           NULL | 2014-01-27 |
| Alexis       | 4100.00 |           NULL | 2015-02-20 |
| Julia        | 3400.00 |           NULL | 2016-06-24 |
| Anthony      | 3000.00 |           NULL | 2017-02-07 |
| Kelly        | 3800.00 |           NULL | 2015-06-14 |
| Jennifer     | 3600.00 |           NULL | 2015-08-13 |
| Timothy      | 2900.00 |           NULL | 2016-07-11 |
| Randall      | 2500.00 |           NULL | 2017-12-19 |
| Sarah        | 4000.00 |           NULL | 2014-02-04 |
| Britney      | 3900.00 |           NULL | 2015-03-03 |
| Samuel       | 3200.00 |           NULL | 2016-07-01 |
| Vance        | 2800.00 |           NULL | 2017-03-17 |
| Alana        | 3100.00 |           NULL | 2016-04-24 |
| Kevin        | 3000.00 |           NULL | 2016-05-23 |
| Donald       | 2600.00 |           NULL | 2017-06-21 |
| Douglas      | 2600.00 |           NULL | 2018-01-13 |
| Jennifer     | 4400.00 |           NULL | 2013-09-17 |
| Pat          | 6000.00 |           NULL | 2015-08-17 |
| Susan        | 6500.00 |           NULL | 2012-06-07 |
| William      | 8300.00 |           NULL | 2012-06-07 |
+--------------+---------+----------------+------------+
38 rows in set (0.00 sec)

D3_96509_Aditya>
```

7. Display employees where the first name or last name starts with S.

```
D3_96509_Aditya> SELECT FIRST_NAME,LAST_NAME FROM employees WHERE FIRST_N
+------------+-----------+
| FIRST_NAME | LAST_NAME |
+------------+-----------+
| Sundar     | Ande      |
| Shelli     | Baida     |
| Sarah      | Bell      |
| Shelley    | Higgins   |
| Susan      | Jacobs    |
| Steven     | King      |
| Sundita    | Kumar     |
| Steven     | Markle    |
| Samuel     | McLeod    |
| Nandita    | Sarchand  |
| Ismael     | Sciarra   |
| John       | Seo       |
| Sarath     | Sewall    |
| John       | Singh     |
| Lindsey    | Smith     |
| William    | Smith     |
| Stephen    | Stiles    |
| Martha     | Sullivan  |
| Patrick    | Sully     |
| Sigal      | Tobias    |
| Sean       | Tucker    |
| Shanta     | Vollman   |
+------------+-----------+
22 rows in set (0.00 sec)

D3_96509_Aditya>
```

8. Display details of jobs in the descending order of the title.

```
D3_96509_Aditya> SELECT *
    -> FROM jobs
    -> ORDER BY job_title DESC;
+------------+---------------------------------+------------+------------+
| JOB_ID     | JOB_TITLE                       | MIN_SALARY | MAX_SALARY |
+------------+---------------------------------+------------+------------+
| ST_MAN     | Stock Manager                   |       5500 |       8500 |
| ST_CLERK   | Stock Clerk                     |       2008 |       5000 |
| SH_CLERK   | Shipping Clerk                  |       2500 |       5500 |
| SA_REP     | Sales Representative            |       6000 |      12008 |
| SA_MAN     | Sales Manager                   |      10000 |      20080 |
| PU_MAN     | Purchasing Manager              |       8000 |      15000 |
| PU_CLERK   | Purchasing Clerk                |       2500 |       5500 |
| PR_REP     | Public Relations Representative |       4500 |      10500 |
| AC_ACCOUNT | Public Accountant               |       4200 |       9000 |
| IT_PROG    | Programmer                      |       4000 |      10000 |
| AD_PRES    | President                       |      20080 |      40000 |
| MK_REP     | Marketing Representative        |       4000 |       9000 |
| MK_MAN     | Marketing Manager               |       9000 |      15000 |
| HR_REP     | Human Resources Representative  |       4000 |       9000 |
| FI_MGR     | Finance Manager                 |       8200 |      16000 |
| AD_VP      | Administration Vice President   |      15000 |      30000 |
| AD_ASST    | Administration Assistant        |       3000 |       6000 |
| AC_MGR     | Accounting Manager              |       8200 |      16000 |
| FI_ACCOUNT | Accountant                      |       4200 |       9000 |
+------------+---------------------------------+------------+------------+
19 rows in set (0.00 sec)
```

9. Display the details of the employees from department 30, earning the salary greater 10000 but do not receive the commission.

```
D3_96509_Aditya> SELECT * FROM employees WHERE DEPARTMENT_ID =30 AND SALARY > 10000 AND COMMISSION_PCT IS NULL;
+-------------+------------+-----------+-------+---------------+------------+--------+----------+----------------+------------+---------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME | EMAIL | PHONE_NUMBER  | HIRE_DATE  | JOB_ID | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-----------+-------+---------------+------------+--------+----------+----------------+------------+---------------+
|         114 | Den        | Li        | DLI   | 1.515.555.0114 | 2012-12-07 | PU_MAN | 11000.00 |           NULL |        100 |            30 |
+-------------+------------+-----------+-------+---------------+------------+--------+----------+----------------+------------+---------------+
1 row in set (0.00 sec)

D3_96509_Aditya>
```

10. Display the details of employees reporting to the managers with employee id 100 and 120

```
D3_96509_Aditya> SELECT * FROM employees WHERE MANAGER_ID IN (100,120);
+-------------+------------+-------------+----------+---------------+------------+----------+----------+----------------+------------+---------------+
| EMPLOYEE_ID | FIRST_NAME | LAST_NAME   | EMAIL    | PHONE_NUMBER  | HIRE_DATE  | JOB_ID   | SALARY   | COMMISSION_PCT | MANAGER_ID | DEPARTMENT_ID |
+-------------+------------+-------------+----------+---------------+------------+----------+----------+----------------+------------+---------------+
|         101 | Neena      | Yang        | NYANG    | 1.515.555.0101 | 2015-09-21 | AD_VP    | 17000.00 |           NULL |        100 |            90 |
|         102 | Lex        | Garcia      | LGARCIA  | 1.515.555.0102 | 2011-01-13 | AD_VP    | 17000.00 |           NULL |        100 |            90 |
|         114 | Den        | Li          | DLI      | 1.515.555.0114 | 2012-12-07 | PU_MAN   | 11000.00 |           NULL |        100 |            30 |
|         120 | Matthew    | Weiss       | MWEISS   | 1.650.555.0120 | 2014-07-18 | ST_MAN   |  8000.00 |           NULL |        100 |            50 |
|         121 | Adam       | Fripp       | AFRIPP   | 1.650.555.0121 | 2015-04-10 | ST_MAN   |  8200.00 |           NULL |        100 |            50 |
|         122 | Payam      | Kaufling    | PKAUFLIN | 1.650.555.0122 | 2013-05-01 | ST_MAN   |  7900.00 |           NULL |        100 |            50 |
|         123 | Shanta     | Vollman     | SVOLLMAN | 1.650.555.0123 | 2015-10-10 | ST_MAN   |  6500.00 |           NULL |        100 |            50 |
|         124 | Kevin      | Mourgos     | KMOURGOS | 1.650.555.0124 | 2017-11-16 | ST_MAN   |  5800.00 |           NULL |        100 |            50 |
|         145 | John       | Singh       | JSINGH   | 44.1632.960000 | 2014-10-01 | SA_MAN   | 14000.00 |           0.40 |        100 |            80 |
|         146 | Karen      | Partners    | KPARTNER | 44.1632.960001 | 2015-01-05 | SA_MAN   | 13500.00 |           0.30 |        100 |            80 |
|         147 | Alberto    | Errazuriz   | AERRAZUR | 44.1632.960002 | 2015-03-10 | SA_MAN   | 12000.00 |           0.30 |        100 |            80 |
|         148 | Gerald     | Cambrault   | GCAMBRAU | 44.1632.960003 | 2017-10-15 | SA_MAN   | 11000.00 |           0.30 |        100 |            80 |
|         149 | Eleni      | Zlotkey     | EZLOTKEY | 44.1632.960004 | 2018-01-29 | SA_MAN   | 10500.00 |           0.20 |        100 |            80 |
|         201 | Michael    | Martinez    | MMARTINE | 1.515.555.0166 | 2014-02-17 | MK_MAN   | 13000.00 |           NULL |        100 |            20 |
|         125 | Julia      | Nayer       | JNAYER   | 1.650.555.0125 | 2015-07-16 | ST_CLERK |  3200.00 |           NULL |        120 |            50 |
|         126 | Irene      | Mikkilineni | IMIKKILI | 1.650.555.0126 | 2016-09-28 | ST_CLERK |  2700.00 |           NULL |        120 |            50 |
|         127 | James      | Landry      | JLANDRY  | 1.650.555.0127 | 2017-01-14 | ST_CLERK |  2400.00 |           NULL |        120 |            50 |
|         128 | Steven     | Markle      | SMARKLE  | 1.650.555.0128 | 2018-03-08 | ST_CLERK |  2200.00 |           NULL |        120 |            50 |
|         180 | Winston    | Taylor      | WTAYLOR  | 1.650.555.0145 | 2016-01-24 | SH_CLERK |  3200.00 |           NULL |        120 |            50 |
|         181 | Jean       | Fleaur      | JFLEAUR  | 1.650.555.0146 | 2016-02-23 | SH_CLERK |  3100.00 |           NULL |        120 |            50 |
|         182 | Martha     | Sullivan    | MSULLIVA | 1.650.555.0147 | 2017-06-21 | SH_CLERK |  2500.00 |           NULL |        120 |            50 |
|         183 | Girard     | Geoni       | GGEONI   | 1.650.555.0148 | 2018-02-03 | SH_CLERK |  2800.00 |           NULL |        120 |            50 |
+-------------+------------+-------------+----------+---------------+------------+----------+----------+----------------+------------+---------------+
22 rows in set (0.00 sec)

D3_96509_Aditya>
```

11. Display the unique country_id from locations table.

```
D3_96509_Aditya> SELECT DISTINCT COUNTRY_ID FROM location;
ERROR 1146 (42S02): Table 'hr.location' doesn't exist
D3_96509_Aditya> SELECT DISTINCT COUNTRY_ID FROM locations;
+------------+
| COUNTRY_ID |
+------------+
| AU         |
| BR         |
| CA         |
| CH         |
| CN         |
| DE         |
| GB         |
| IN         |
| IT         |
| JP         |
| MX         |
| NL         |
| SG         |
| US         |
+------------+
14 rows in set (0.00 sec)
```

12. Display all employees whose have the job_id as IT_PROG and FI_ACCOUNT.

13. Display all countries in ascending order.

```
D3_96509_Aditya> SELECT * FROM countries ORDER BY COUNTRY_NAME ASC;
+------------+----------------------------------------+-----------+
| COUNTRY_ID | COUNTRY_NAME                           | REGION_ID |
+------------+----------------------------------------+-----------+
| AR         | Argentina                              |        20 |
| AU         | Australia                              |        40 |
| BE         | Belgium                                |        10 |
| BR         | Brazil                                 |        20 |
| CA         | Canada                                 |        20 |
| CN         | China                                  |        30 |
| DK         | Denmark                                |        10 |
| EG         | Egypt                                  |        50 |
| FR         | France                                 |        10 |
| DE         | Germany                                |        10 |
| IN         | India                                  |        30 |
| IL         | Israel                                 |        30 |
| IT         | Italy                                  |        10 |
| JP         | Japan                                  |        30 |
| KW         | Kuwait                                 |        30 |
| ML         | Malaysia                               |        30 |
| MX         | Mexico                                 |        20 |
| NL         | Netherlands                            |        10 |
| NG         | Nigeria                                |        50 |
| SG         | Singapore                              |        30 |
| CH         | Switzerland                            |        10 |
| GB         | United Kingdom of Great Britain and Nort |        10 |
| US         | United States of America               |        20 |
| ZM         | Zambia                                 |        50 |
| ZW         | Zimbabwe                               |        50 |
+------------+----------------------------------------+-----------+
25 rows in set (0.00 sec)
```